



# Zherish Galvin P. Mayordo

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## SUMMARY

Computer Engineering graduate specializing in AI-powered business automation, intelligent workflow orchestration, and system integrations. Experienced in designing and developing AI agents, Retrieval-Augmented Generation (RAG) pipelines, and API-driven automation solutions that streamline business operations and reduce manual effort. Proficient in Python, JavaScript, n8n, REST APIs, Docker, Linux, and cloud infrastructure, with a strong foundation in software engineering and scalable system design.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript/JavaScript, SQL, Bash

**AI Engineering:** AI Agents, RAG, LLM Integration, Prompt Engineering, Vector Databases, MCP

**Automation:** n8n, Workflow Orchestration, REST APIs, Webhooks, OAuth 2.0

**Backend:** FastAPI, Flask, Fastify, Express.js, Next.js

**Cloud & DevOps:** Docker, GCP, GitHub Actions, Git, Linux

**Integrations:** Google Workspace APIs, Meta Graph API, Telegram Bot API, OpenAI API, Gemini API, OpenRouter

## CAREER HISTORY

### Trading Algorithm Programmer

Jul 2026 – Present

Robin Ho

Singapore (Remote)

- Developing automations and trading algorithms using Pine Script and MQL5 to help traders identify and execute market opportunities.
- Analyzing market conditions, price patterns, and economic events to design and improve algorithmic trading strategies.

### AI Automation Specialist

Jun 2026 – Present

Freelance

Quezon City, PH

- Developing automation solutions using n8n for service-based businesses to increase operational efficiency.
- Designed and developed AI-powered automation workflows using n8n, integrating LLMs, REST APIs, webhooks, and third-party services to streamline business operations.
- Built AI solutions for lead generation, lead nurturing, email automation, client onboarding, document processing, and internal business processes.
- Developed Retrieval-Augmented Generation (RAG) pipelines, AI agents, and prompt-driven workflows for context-aware information retrieval and response generation.
- Integrated external platforms using REST APIs, OAuth 2.0, Google Workspace APIs, Meta Graph API, and Telegram Bot API.
- Deployed and maintained self-hosted n8n infrastructure on Google Cloud Platform using Cloud Run, PostgreSQL, and Docker, implementing secure, scalable, and reliable deployments.
- Designed modular workflows with robust error handling, retry mechanisms, logging, and human-in-the-loop approval processes to improve reliability and maintainability.
- Collaborated with clients to analyze business requirements and deliver scalable AI-powered automation solutions tailored to their workflows.

### Professional Development

Nov 2025 – May 2026

Full-Time AI & Software Engineering Study

- Pursued full-time study to transition from software engineering into AI application development.
- Completed the **Machine Learning Specialization** (DeepLearning.AI & Stanford University) and the **IBM DevOps and Software Engineering Professional Certificate**, strengthening expertise in machine learning, cloud technologies, DevOps, and software engineering.

- Designed and built AI projects involving Retrieval-Augmented Generation (RAG), LLM integration, API development, and backend automation to gain hands-on experience with production-oriented AI systems.
- Established the technical foundation that led to working as an AI Automation Specialist focused on building AI-powered business solutions.

## Trading Algorithm Programmer

Jul 2024 – Nov 2025

Robin Ho

Singapore (Remote)

- Developed algorithmic trading strategies and technical indicators based on employer specifications using TradingView (Pine Script) and MetaTrader 5 (MQL5)
- Deployed trading algorithms and technical indicators across multiple asset classes, including stock indices, foreign exchange (forex), commodities, and cryptocurrencies remotely on Windows Virtual Private Server (VPS)
- Built a Telegram chatbot leveraging webhooks and Telegram BotFather to automate signal delivery for subscribed users

## Researcher Intern

Mar 2024 – Jul 2024

DOST - Advanced Science and Technology Institute

Quezon City, PH

- Trained a convolutional neural network (CNN) model in TensorFlow for motor condition diagnosis on 3,750 time-series motor current signals transformed into STFT, Gabor, and Wavelet representations
- Experimented with different configurations of STFT, Gabor, and Wavelet transformations to achieve the best results for the CNN model
- Performed exploratory data analysis (EDA) on publicly available predictive maintenance datasets on motor condition diagnosis
- Co-authored IEEE conference papers on deep learning-based fault diagnosis using signal transformation techniques

## EDUCATIONAL LEVEL

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### Technological Institute of the Philippines

Quezon City, PH

Bachelor of Science in Computer Engineering, Specialization in Intelligent Systems

Aug 2020 – May 2025

- President's Lister: 2nd Semester A.Y. 2020-2021, 1st Semester A.Y. 2022-2023
- VPAA's Lister: 1st Semester A.Y. 2021-2022
- Dean's Lister: 1st Semester A.Y. 2023-2024
- Software Design and Architecture, Software Development Lifecycle (SDLC), Linux, Data Structures and Algorithms, Cisco Networking

## TRAININGS & CERTIFICATIONS

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*Note: Certifications include embedded links for verification*

### DevOps and Software Engineering Professional Certificate [VIEW](#)

Nov 2025 - Mar 2026

IBM

DevOps, Cloud, Software Engineering

- Covered SDLC, Agile methodologies, microservices and serverless architectures, TDD/BDD, and core DevOps practices including CI/CD, containerization, container orchestration, application security, and observability
- Implemented CI/CD pipelines and DevOps workflows using GitHub Actions, Docker, Kubernetes/OpenShift, Tekton, Snyk, and Prometheus through hands-on labs

### Machine Learning Specialization [VIEW](#)

Oct 2025 - Nov 2025

DeepLearning.AI & Stanford University

Python, Machine Learning, Model Evaluation

- Completed supervised, unsupervised, and reinforcement learning modules covering key algorithms such as linear regression, clustering, and Q-learning using Python and TensorFlow
- Applied best practices in building, training, and evaluating machine learning models, including regularization, bias-variance tradeoff, and performance metrics
- Implemented end-to-end machine learning pipelines to solve real-world problems across classification, regression, and recommendation tasks

## ACHIEVEMENTS & OTHER EXPERIENCES

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*Note: Some project, achievement, and research titles include embedded links to supporting resources*

### AI Facebook Messenger Assistant

2026

n8n, AI, RAG, Meta Graph API



- Built an AI-powered Facebook Messenger assistant that automates customer inquiries using Retrieval-Augmented Generation (RAG), conversation memory, and a business knowledge base.
- Designed a multi-stage retrieval and answer verification pipeline that decomposes complex queries, validates responses against retrieved context, and requests clarification when sufficient information is unavailable.

### PDF/TXT to ProPresenter-ready Lyrics Formatter

2026

n8n, AI, Automation



- Built an AI-powered automation workflow for a church that extracts, cleans, and formats lyrics from PDF and TXT files into a ProPresenter-ready format for direct import.
- Reduced lyrics preparation time from 5–10 minutes to a few seconds while minimizing formatting errors and eliminating repetitive manual work.

### Smart Reddit Search

2026

n8n, AI, RAG, Automation



- Built an automated AI-powered Reddit search workflow that retrieves, filters, and indexes Reddit posts into a vector database for semantic search.
- Reduced manual searching and skimming of Reddit discussions from several minutes to a few seconds, enabling faster research and information retrieval.

### Design of Optimized Path Generation for Physical Guidance-Driven Collaborative Robot Using Metaheuristic Algorithm

2025

Python, FastAPI, JavaScript, React, Next.js, C++, Arduino



- Built an end-to-end hardware and software system for a collaborative robotic arm, allowing users to control the robot arm using their mobile device
- Automated system setup using Bash and Python scripting to initialize required services and establish WebSocket connections, reducing setup to WiFi configuration only and enabling immediate operation
- Integrated Learning from Demonstration (LfD) with Genetic Algorithm (GA)-based optimization to improve task execution of the robotic arm
- Reduced execution time by over 90% while maintaining full task accuracy

### Project MaSense – Motor Fault Diagnosis System for Predictive Maintenance

2024

Python, TensorFlow, Data Engineering, Model Evaluation

- Co-authored "Machine Condition Diagnosis Using Deep Learning and Gabor-Transformed Motor Current Signatures" published in TENCON 2024 - 2024 IEEE Region 10 Conference (TENCON)  
— [ieeexplore.ieee.org/document/10903100](https://ieeexplore.ieee.org/document/10903100)
- Co-authored "Deep Learning-Based Machine Condition Diagnosis Using Short-Time Fourier Transformation Variants" published in 2024 International Conference on Diagnostics in Electrical Engineering (Diagnostika)  
— [ieeexplore.ieee.org/document/10693710](https://ieeexplore.ieee.org/document/10693710)